Final Design Review For Mounting Brace

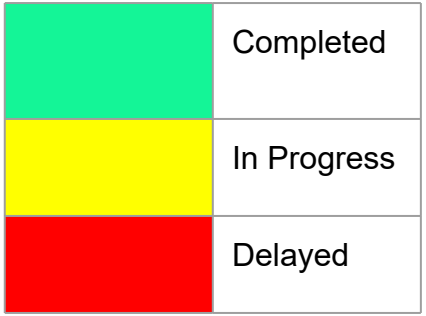
By Angie Foo, Brett Beaudoin, Jaylan Mo, Joshan Darji

Team Member Roles and Responsibilities

Name	Role and Responsibilities
Angie Foo	Design validation + qualification + cost analysis
Brett Beaudoin	Prototype on FDM
Jaylan Mo	CAD modeling
Joshan Darji	Part Manufacturing on SLM + Post Processing

Schedule

Task	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
CAD Design/ Drawings						
PDR						
Prototype on FDM						
Critical Design Review						
Part manufacturing on SLM +Post Processing						
Qualification + Cost analysis						
Final Presentation						



Risk Assessment Scorecard

Proposal	Issue	Strategy/Plan	Owner	Due Date	Technical Risk	Schedule
1	Support generation complexity, potential warping or print failure	Evaluate multiple support strategies (orientation changes, custom supports, lattice supports). Run print simulation to predict distortion	Angie	Week 7	3	3
2	Warpage, dimensional accuracy at mounting holes, support removal inside lower lattice	Optimize print orientation, add radius at bend, print undersized and post machine holes, adjust lattice angle	Buddy	Week 7	2	2

Design Proposal #1: Generative Design

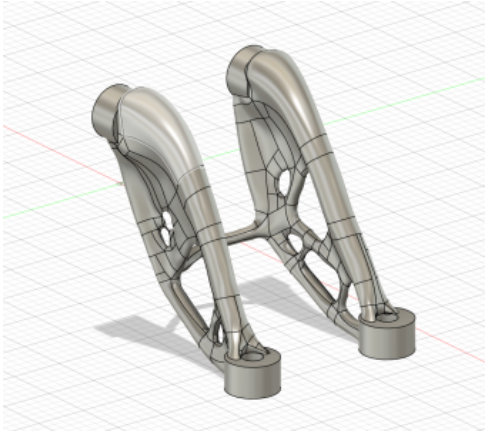
Material: Stainless Steel 316L

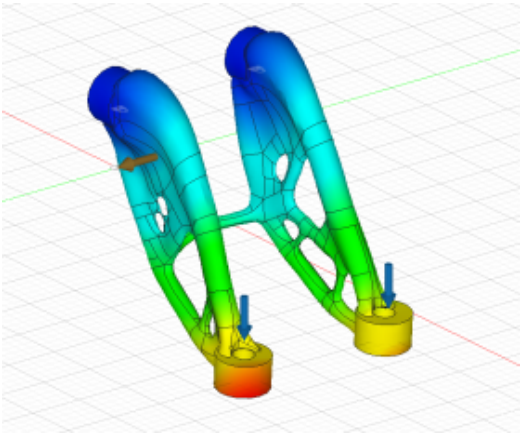
Weight Reduction:

- **Original Mass:** 25.733 grams
- **Optimized Mass:** 6.984 grams
- **Mass Reduction:** 72.86%

Structural Performance:

- **Max displacement:** 0.019 mm



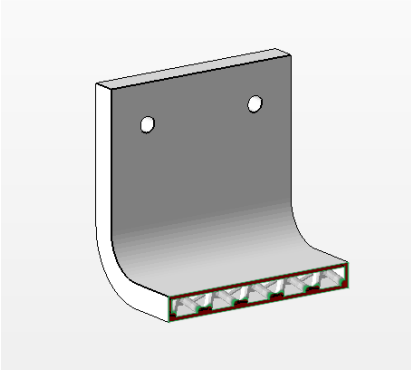


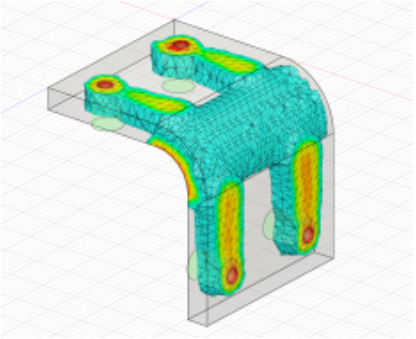
Design Proposal #2: Lattice

Material: Stainless Steel 316L

Weight Reduction:

- **Original Mass**: 25.733 grams
- **Optimized Mass**: 11.984g
- **Mass Reduction:** 53.43%





Structural Performance:

- Unable to calculate solid body max displacement in Fusion 360, in theory should have smaller max displacement than Design Proposal 1

Outside structure is the same as original, interior has lattice structure (sliced picture on left)

Hole will be drilled in between original bracket holes for powder to be emptied out (forces are less generated in that area)

Design Proposal #3: Shape Optimization

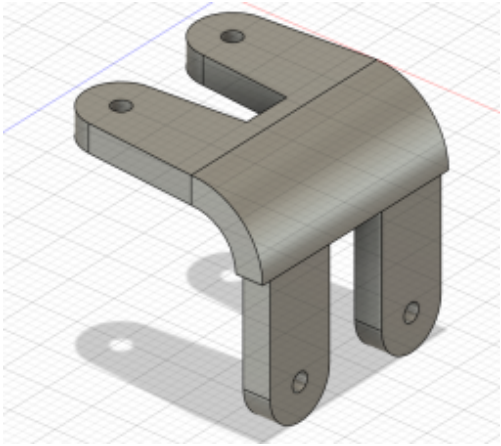
Material: Stainless Steel 316L

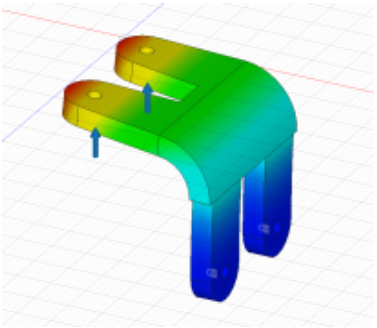
Weight Reduction:

- **Original Mass**: 25.733 grams
- **Optimized Mass**: 17.931 grams
- **Mass Reduction:** 30.31%

Structural Performance:

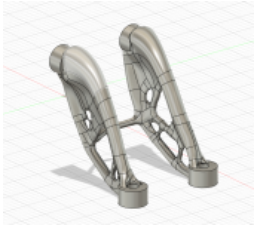
- **Max displacement:** 0.048 mm



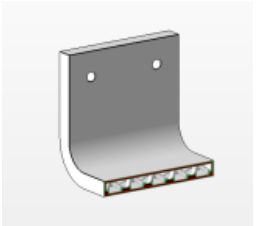


Comparison of Design Options

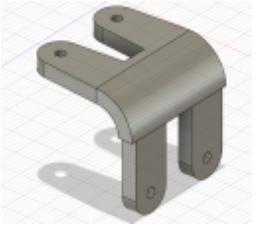
- Design Proposal #1: Lighter in weight and volume, takes less time to manufacture through additive manufacturing
- Design Proposal #2: Simpler in design, user friendly, however inability to determine FEA and no real way to empty powder after production eliminates this proposal
- Design Proposal #3: Simple design, more user friendly and familiar geometry with original part, heavier part than other 2 proposals.
- Decide to choose design proposal #1 since uses less material and is more resilient to deformation



#1



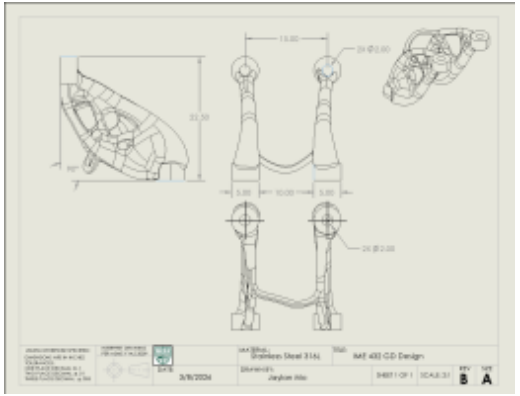
#2



#3

Final Bracket Design and Qualification





Feature Name	Nominal value	Tolerance	Measured Value	Within Spec (Y/N)
Total Bracket Bend Angle	90 degrees	1 degree	93.8 degrees	N
Distance Between Outer Diameter of Holes	10 mm	0.3 mm	10.964 mm	N
Bottom Right Hole Diameter	2 mm	0.3 mm	1.736 mm	Y
Bottom Left Hole Diameter	2 mm	0.3 mm	1.865 mm	Y
Distance between Center of Holes	15 mm	0.3 mm	15.913	N

Cost Analysis

Material Cost for Stainless Steel 316L: $35/kg
Machine Cost (Residual after 5 years): $100,000
Labor Cost: $25/hr
Total Print Time: 3 hours

Total Material Cost: 0.00749*35= $0.26
Labor Cost: 25*3=$75
Cost/part: $75.26
Total Cost+Machine: 100,075.26

Given		
Material Cost	$ 35.00	
Machine Cost	$ 500,000.00	
Labor cost	$ 25.00	hr
Residual cost of the machine	100000	
Ours		
Layer	1217	
Bracket Weight	0.00749	kg
Total Time	3	hr
Labor cost	$ 75.00	
Material Cost	$ 0.26	
Machine Cost	$ 100,000.00	
Total cost to produce part	$ 100,075.26	

Key Learnings

Teams Achievement:

We designed three improved bracket designs and decided on the Generative Design for the bracket. We were able to use the SLM machine to create the bracket out of Stainless Steel 316L.

What went well?

- Successfully created the part with the SLM without major design errors or failure
- Reduced product weight and stayed within max displacement requirements
- Post processing provided a comfortable surface finish and support removal did not damage product

What could be better?

- Part geometry could be improved, some measurements not within tolerance
- Improve geometry design, connecting segment needs to be thicker/sturdier
- The first design

Next Steps

- Redesign the bracket to have a stronger middle brace to resist deformation in distances between the holes of the part
- Calibrate machine and improve post processing to be within tolerances

